

Multiparameter Case: Estimation

- Let $X_1, \dots, X_n \stackrel{iid}{\sim} f(x; \theta)$, where $\theta = (\theta_1, \dots, \theta_p) \in \mathbb{R}^p$
- Likelihood: $L(\theta) = \prod_{i=1}^n f(X_i; \theta)$
- Log-likelihood: $\ell(\theta) = \sum_{i=1}^n \log f(X_i; \theta)$
- Score equation:

$$\frac{\partial \ell(\theta)}{\partial \theta_j} = 0, \quad j = 1, \dots, p$$

- MLE: $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_p)$

An Example

- Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, where $\theta = (\mu, \sigma) \in \mathbb{R}^2$
- Likelihood:

$$L(\theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} = \frac{1}{(2\pi)^{n/2} \sigma^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

- Log likelihood:

$$\ell(\mu, \sigma) = -\frac{n}{2} \log 2\pi - n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

- Score equations:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0$$

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

- MLE:

$$\hat{\mu} = \bar{X}$$

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- MLE for variance:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

- Sample variance:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- MLE is NOT sample variance; sample variance is unbiased; both are consistent.

Fisher Information Matrix

- Gradient:

$$\nabla \log f(x; \theta) = \left(\frac{\partial \log f(x; \theta)}{\partial \theta_1}, \dots, \frac{\partial \log f(x; \theta)}{\partial \theta_p} \right)' \in \mathbb{R}^p$$

- Fisher information matrix:

$$I(\theta) = \text{Cov}(\nabla \log f(x; \theta)) \in \mathbb{R}^{p \times p}$$

- Elements of $I(\theta)$:

$$\begin{aligned} I_{jk} &= \text{Cov}\left(\frac{\partial \log f(x; \theta)}{\partial \theta_j}, \frac{\partial \log f(x; \theta)}{\partial \theta_k}\right) \\ &= \mathbb{E}\left(\frac{\partial \log f(x; \theta)}{\partial \theta_j} \frac{\partial \log f(x; \theta)}{\partial \theta_k}\right) \\ &= -\mathbb{E}\left(\frac{\partial^2 \log f(x; \theta)}{\partial \theta_j \partial \theta_k}\right) \end{aligned}$$

- $I(\theta)$ is a $p \times p$ symmetric and positive definite matrix
- Theorem: If $Y = u(X_1, \dots, X_n)$ is an unbiased estimator of θ_j . Then,

$$\text{Var}(Y) \geq \frac{1}{n} \left(I^{-1}(\theta) \right)_{jj}$$

An Example

- $X \sim N(\mu, \sigma^2)$

$$\log f = -\frac{1}{2} \log 2\pi - \log \sigma - \frac{1}{2\sigma^2} (x - \mu)^2$$

$$\frac{\partial \log f}{\partial \mu} = \frac{1}{\sigma^2} (x - \mu)$$

$$\frac{\partial \log f}{\partial \sigma} = -\frac{1}{\sigma} + \frac{1}{\sigma^2} (x - \mu)^2$$

$$\frac{\partial^2 \log f}{\partial \mu^2} = -\frac{1}{\sigma^2}$$

$$\frac{\partial^2 \log f}{\partial \sigma^2} = \frac{1}{\sigma^3} - \frac{3}{\sigma^4} (x - \mu)^2$$

$$\frac{\partial^2 \log f}{\partial \mu \partial \sigma} = -\frac{2}{\sigma^3} (x - \mu)$$

- Fisher information matrix:

$$I(\theta) = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{bmatrix}$$

- Inverse:

$$I^{-1}(\theta) = \begin{bmatrix} \sigma^2 & 0 \\ 0 & \frac{\sigma^2}{2} \end{bmatrix}$$

- If Y is an unbiased estimator of μ based on n samples,

$$\text{var}(Y) \geq \frac{\sigma^2}{n}$$

- Check $\bar{X}$:

$$\mathbb{E}\bar{X} = \mu, \quad \text{var}(\bar{X}) = \frac{\sigma^2}{n}$$

- Theorem: Let $X_1, \dots, X_n \stackrel{iid}{\sim} f(x; \theta)$, where $\theta = (\theta_1, \dots, \theta_p)'$. Under some regularity conditions:
 1. The MLE $\hat{\theta}$ is obtained by solving $\frac{\partial \ell(\theta)}{\partial \theta} = 0$, and $\hat{\theta} \xrightarrow{P} \theta$.
 - 2.

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{D} N(0, I^{-1}(\theta))$$